

Step 7: Report and Documentation

BCS3210 Blockchains group project: Decentralized Digital Identity and Data Sharing Platform

What we did in this step

We wrote the final report, made the presentation, and documented the code so that someone else can run everything on a local Hardhat network.

Deliverable	File
Final report (coursebook format)	deliverables/report.pdf
Presentation (12 slides)	deliverables/presentation.pdf
Step deliverables 1-7	deliverables/stepN-deliverable.pdf
Code with run instructions	digital-identity-platform/ and its README.md
Raw measurements	results/ (test output, gas reports, simulation JSON, plot)

1. Report

The report follows the structure and the format rules of the coursebook:

Coursebook requirement	How we did it
Cover page with title, names and IDs	Page 1 (names and IDs still to be filled in)
Introduction, Architecture, Implementation, Experimental Results, Discussion and Conclusion	Sections 1 to 5
No implementation details in the Architecture section	Section 2 only describes roles, data, components and flows
Max 10 pages without references and code	10 pages including cover and references
10 pt Times New Roman, double spaced, A4, 1 inch margins, page numbers	Set in the report stylesheet; page numbers added when the PDF is built
Consistent, verifiable references	Numbered references [1] to [10]
Contribution of each member, where and how AI was used	Section 6 with a transparency statement (to be completed by the team)

2. Project brief checklist

Requirement from the brief	Where
Problem statement, roles, functional requirements, interaction overview	Step 1, report 1 and 2.1
Attribute table (on-chain / off-chain / hashed)	Step 2, report 2.2
Consent model: data types, 1-365 days, who grants/revokes, expiry, workflows	Step 2, report 2.4
Audit log: who, what, when, granted/denied, not deletable	Step 2, report 2.5, DataSharing.sol
Architecture diagram	Step 2, report Figure 1

Requirement from the brief	Where
Register user, retrieve info, set consent (with token reward), revoke consent, log access	DigitalIdentity.sol, ConsentManager.sol, DataSharing.sol
Data stays with the user, only hashes on-chain, access only with valid consent	Python vault (offchain/), report 2.5
Tokens reward sharing, only the contract can mint, no tokens move during access	AccessToken.sol, tests test_RevertsWhenNonMinterMints, test_NoTokensMoveDuringAccess
Solidity unit tests named after the contracts, why each is critical	test/*.t.sol (64 passing), Step 4
Gas table: deployment and average per function, optimisation	Step 4, report 4.2
Integration tests of the full workflow	Integration.t.sol, offchain/demo.py
No personal data in tests	Generated addresses, fake hashes, generated ID images
Local deployment with Hardhat	Ignition module, npm run deploy:local
Simulation with several roles, gas and time, scaling tables	Step 5, report 4.3
Front-end with Viem.js on the local network (bonus)	Step 6, frontend/
Instructions to compile, deploy and test	README.md

3. How to reproduce everything

```

cd digital-identity-platform
npm install
npx hardhat compile
npx hardhat test           # 64 tests
npm run gas-report         # gas per function
npm run demo               # walk-through
npx hardhat node           # second terminal
npm run deploy:local
npm run simulate           # scaling results
python offchain/make_sample_data.py && python offchain/demo.py
npm run frontend           # http://localhost:5173
python deliverables/build_pdf.py           # rebuild step PDFs
python deliverables/build_pdf.py    report slides

```

We tested these steps on a fresh copy of the project (only the source files, `npm install` from scratch) before submitting.

4. Still to do by the team before the deadline

- Fill in names and student IDs on the cover page, the contribution table and the title slide.
- Check and adjust the AI transparency statement in Section 6 of the report.
- Every member reads the code and the report so they can explain it.
- Submit the project before **Sunday 4 October 2026, 23:59 CEST** and send the presentation to the instructors before **12:00 on Monday 5 October**.